

Scalability & Future Plan

Date	3 July 2024
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-machine Flask deployment.	Performance may decrease with increasing user requests.	High
2	The application does not include user authentication or role-based access control.	Unauthorized users can access the system.	Medium
3	The system provides only approval/rejection results without detailed explanations.	Users may not understand the reasons behind the prediction.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a local Flask server.	Deploy the application on cloud platforms (AWS, Azure, or GCP) with load balancing and auto-scaling.
2	Data Storage	Uses a CSV file (loan_prediction.csv) for training and does not store user requests.	Integrate a relational database such as MySQL or PostgreSQL to manage user data and prediction history.
3	Performance	Prediction performance is suitable for small-scale usage with a pre-trained XGBoost model.	Implement caching, API optimization, and containerization (Docker) to improve response time and support larger workloads.

4	Security	No user authentication, authorization, or data encryption mechanisms are implemented.	Add user login, role-based access control, HTTPS encryption, and secure data handling practices.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase1	Integrate a MySQL database to store user applications and prediction history.	1–2 Months	Improved data management and user record maintenance.
Phase2	Implement user authentication and role-based access control.	2–3 Months	Enhanced security and controlled system access.
Phase3	Deploy the application on a cloud platform (AWS/Azure/GCP).	Next 4 Months	Increased accessibility, scalability, and reliability.
Phase4	Add Explainable AI (XAI) features to justify loan approval decisions.	Future Release	Greater transparency and user trust in predictions.